

How many outcomes are in each sample space?

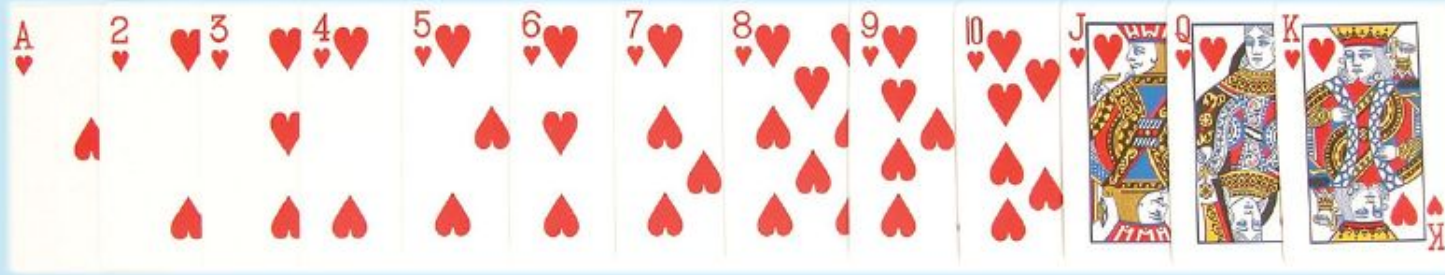
- a** A number between 4 and 7 is chosen at random.
- b** A ticket is chosen from tickets numbered 1 to 99.
- c** A ticket is chosen from tickets numbered 0 to 100.
- d** Ten cards are placed in a hat and one is chosen at random.

Liam throws a dice. Write, as a fraction, the probability that he will throw:

- | | |
|--------------------------------|-------------------------------|
| a a 2 | b a zero |
| c an even number | d an odd number |
| e either a 1 or a 6 | f a number less than 4 |
| g a number less than 7. | |



A standard pack of cards has 4 suits: hearts (♥), diamonds (♦), clubs (♣) and spades (♠). There are 13 cards in each suit and 52 cards altogether.



The first of these cards is called an **Ace**. The last three cards are the *picture* or *court* cards: the **Jack**, the **Queen** and the **King**.

From the 13 cards shown above, one card is to be chosen at random. What is the probability that the card chosen will be:

- | | | |
|-------------------|-------------------|------------------------------|
| a the 7? | b the Ace? | c a picture card? |
| d a heart? | e a spade? | f either a 9 or a 10? |

From a standard pack of 52 cards, a card is chosen at random. What is the probability that the card will be:

- | | | |
|---------------------|--------------------------|-----------------------------------|
| a a diamond? | b a red card? | c a black card? |
| d a 3? | e a picture card? | f either an Ace or a King? |